

Urban Systems & Product Design · Session-by-Session Reading List

Undergraduate: 1–3 readings per session, selected for clarity and discussion value · Graduate / Master's: 6–10 readings, full theoretical and empirical apparatus

WK	TOPIC & SESSION FOCUS	UNDERGRADUATE (1–3 readings) <i>Synthesis & discussion focus</i>	GRADUATE / MASTER'S (6–10 readings) <i>Full scholarly apparatus</i>
1	Cities as Systems <i>Urban complexity: nodes, flows, feedback loops, and emergence. Why cities resist simple optimization.</i>	- Jacobs, Jane, <i>The Death and Life of Great American Cities, Introduction & Ch. 1</i> (1961) — the foundational text — accessible and urgent - Townsend, Anthony, <i>Smart Cities, Introduction & Ch. 1</i> (2013) — contemporary digital urbanism context - Hoffman, Patrick T., <i>Urban Systems & Product Design — Course Introduction</i> (2026) — @arcktip framing essay	- Jacobs, Jane, <i>The Death and Life of Great American Cities, Introduction & Ch. 1–2</i> (1961) - Townsend, Anthony, <i>Smart Cities, Introduction & Ch. 1</i> (2013) - Batty, Michael, <i>The New Science of Cities, Introduction</i> (2013) — MIT Press — complexity and urban modeling - Portugali, Juval, <i>Self-Organization and the City, Introduction</i> (2000) — Springer — emergence and urban form - Hoffman, Patrick T., <i>Urban Systems & Product Design — Course Introduction</i> (2026) — @arcktip framing essay - Glaeser, Edward, <i>Triumph of the City, Ch. 1</i> (2011) — cities as engines of human productivity - Hall, Peter, <i>Cities of Tomorrow, Introduction</i> (2002) — Blackwell — history of planning thought
2	The Planning Tradition <i>Zoning, master plans, and the limits of top-down design. Why cities resist optimization and planners often make things worse.</i>	- Jacobs, Jane, <i>The Death and Life of Great American Cities, Ch. 22–23</i> (1961) — her critique of orthodox planning — essential - Scott, James C., <i>Seeing Like a State, Introduction & Ch. 1</i> (1998) — why high-modernist planning fails - Bertaud, Alain, <i>Order Without Design, Introduction & Ch. 1</i> (2018) — markets and planning as complements	- Jacobs, Jane, <i>The Death and Life of Great American Cities, Ch. 22–23</i> (1961) - Scott, James C., <i>Seeing Like a State, Introduction & Ch. 1–2</i> (1998) — Yale UP — legibility and state simplification - Bertaud, Alain, <i>Order Without Design, Introduction & Ch. 1–2</i> (2018) — MIT Press — urban economics meets planning practice - Hall, Peter, <i>Cities of Tomorrow, Ch. 3 & 8</i> (2002) — Howard, Le Corbusier, and their legacies - Fischler, Raphaël, <i>The Metropolitan Dimension of Early Zoning</i> (1998) — Journal of the American Planning Association - Talen, Emily, <i>New Urbanism and American Planning, Introduction</i> (2005) — Routledge — critique and reform - Hirt, Sonia, <i>Zoned in the USA, Introduction & Ch. 1</i> (2014) — Cornell UP — American zoning's global uniqueness - Flyvbjerg, Bent, <i>Rationality and Power, Introduction</i> (1998) — University of Chicago Press — planning as politics
3	Data & the City <i>How quantitative methods transformed urban planning.</i>	Mattern, Shannon, <i>A City Is Not a Computer</i> (2017) — Places Journal — free, brief, canonical critique	Mattern, Shannon, <i>A City Is Not a Computer</i> (2017) — Places Journal

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	GIS, dashboards, and the risks of data reductionism.	- Townsend, Anthony, Smart Cities, Ch. 3–4 (2013) — <i>civic hackers and open data</i> - Green, Ben, The Smart Enough City, Introduction (2019) — MIT Press — <i>technology in its proper place</i>	- Townsend, Anthony, Smart Cities, Ch. 3–4 (2013) - Green, Ben, The Smart Enough City, Introduction & Ch. 1 (2019) — MIT Press - Kitchin, Rob, The Data Revolution, Introduction & Ch. 1 (2014) — SAGE — <i>what big data is and isn't</i> - Crampton, Jeremy et al., Beyond the Geotag: Situating 'Big Data' (2013) — <i>Dialogues in Human Geography</i> - O'Neil, Cathy, Weapons of Math Destruction, Introduction & Ch. 1 (2016) — Crown — <i>algorithmic harm in civic contexts</i> - Eubanks, Virginia, Automating Inequality, Introduction (2018) — St. Martin's — <i>automated decision-making and the poor</i> - Loukissas, Yanni, All Data Are Local, Introduction (2019) — MIT Press — <i>data as situated, not universal</i>
4	Mobility Systems I <i>Transit networks, mode share, induced demand, and the political economy of transportation investment.</i>	- Jacobs, Jane, The Death and Life of Great American Cities, Ch. 18 (1961) — <i>erosion of cities by automobiles</i> - Vanderbilt, Tom, Traffic, Introduction & Ch. 1 (2008) — Knopf — <i>psychology and systems of movement</i> - Walker, Jarrett, Human Transit, Introduction & Ch. 1–2 (2011) — Island Press — <i>the geometry of transit</i>	- Jacobs, Jane, The Death and Life of Great American Cities, Ch. 18 (1961) - Vanderbilt, Tom, Traffic, Introduction & Ch. 1 (2008) - Walker, Jarrett, Human Transit, Introduction & Ch. 1–3 (2011) — Island Press — <i>essential transit geometry</i> - Downs, Anthony, Still Stuck in Traffic, Ch. 1–2 (2004) — Brookings — <i>induced demand and congestion</i> - Cervero, Robert, The Transit Metropolis, Introduction & Ch. 1 (1998) — Island Press — <i>transit and urban form globally</i> - Giuliano, Genevieve & Hanson, Susan, The Geography of Urban Transportation, Ch. 1 (2004) — Guilford — <i>comprehensive overview</i> - Levinson, David & Krizek, Kevin, The End of Traffic and the Future of Access, Introduction (2017) — Network Design Lab — <i>digital mobility futures</i> - Litman, Todd, Transportation Cost and Benefit Analysis, Ch. 1 (2024) — Victoria Transport Policy Institute — <i>free online</i>
5	Mobility Systems II <i>Micromobility deep-dive: the Social Bicycles → JUMP → Uber arc as a full product lifecycle and urban policy case study.</i>	- Hoffman, Patrick T., Dockless: What We Learned Building the Bike That Became JUMP (forthcoming) — @arcktip — <i>primary practitioner account, essential</i> - Fishman, Elliot, Bikeshare: A Review of Recent Literature (2016) — Transport Reviews — <i>empirical overview</i> - NACTO, Urban Street Design Guide (Bicycle Facilities chapter) (2013) — <i>professional standard</i> — <i>free online</i>	- Hoffman, Patrick T., Dockless: What We Learned Building the Bike That Became JUMP (forthcoming) — @arcktip — <i>primary practitioner account</i> - Fishman, Elliot, Bikeshare: A Review of Recent Literature (2016) — Transport Reviews - NACTO, Urban Street Design Guide (Bicycle Facilities & Shared Mobility chapters) (2013) - Pucher, John & Buehler, Ralph, City Cycling, Introduction &

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			Ch. 1 (2012) — MIT Press — comparative international evidence - Sheller, Mimi, Mobility Justice, Ch. 1–2 (2018) — Verso — equity and the right to move - Shaheen, Susan & Chan, Nelson, Dockless Bike Sharing (2016) — Transportation Sustainability Research Center - McNeil, Nathan et al., Breaking Barriers to Bike Share (2017) — Transportation Research and Education Center — equity gaps - SFMTA, Powered Scooter Share Pilot Evaluation Report (2019) — primary policy document — real regulatory decision-making
6	Platform Urbanism <i>How Airbnb, Uber, and DoorDash restructure housing, labor, and neighborhood character through platform logic.</i>	- Srnicek, Nick, Platform Capitalism, Ch. 1–2 (2017) — Polity Press — concise essential framing - Wachsmuth, David & Weisler, Alexander, Airbnb and the Rent Gap (2018) — Environment and Planning A — key empirical argument - Rosenblat, Alex, Uberland, Introduction & Ch. 2 (2018) — UC Press — platform labor and space	- Srnicek, Nick, Platform Capitalism, Ch. 1–3 (2017) - Wachsmuth, David & Weisler, Alexander, Airbnb and the Rent Gap (2018) — Environment and Planning A - Rosenblat, Alex, Uberland, Introduction & Ch. 1–3 (2018) - Zuboff, Shoshana, The Age of Surveillance Capitalism, Introduction & Ch. 1 (2019) — platform logic and behavioral data - Graham, Mark & Zook, Matthew, Augmented Realities and Uneven Geographies (2013) — Environment and Planning A - van Doorn, Niels, Platform Labor: On the Gendered and Racialized Exploitation of Low-income Service Work (2017) — Information, Communication & Society - Sadowski, Jathan, Too Smart, Introduction & Ch. 1–2 (2020) — MIT Press — digital capitalism critique - Thrift, Nigel, Knowing Capitalism, Ch. 1 (2005) — SAGE — spatial dimensions of platform economy
7	Field Session <i>Half-day city walk with structured observation brief. Students map mobility flows, friction points, and equity gaps in the field.</i>	- Whyte, William H., The Social Life of Small Urban Spaces, Introduction & Ch. 1 (1980) — essential — observational method in practice - Gehl, Jan, Life Between Buildings, Introduction (1971) — Island Press — human-scale observation	- Whyte, William H., The Social Life of Small Urban Spaces, Introduction & Ch. 1–2 (1980) - Gehl, Jan, Life Between Buildings, Introduction & Ch. 1 (1971) - Lynch, Kevin, The Image of the City, Introduction & Ch. 1 (1960) — MIT Press — mental maps and legibility - Appleyard, Donald, Livable Streets, Ch. 1 (1981) — UC Press — traffic volume and street life - Jacobs, Allan B., Great Streets, Introduction (1993) — MIT Press — comparative street anatomy
8	Equity & Access <i>Who does the city work for?</i>	Rothstein, Richard, The Color of Law, Introduction & Ch. 1	Rothstein, Richard, The Color of Law, Introduction & Ch. 1–2

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	<i>Designing products that expand access rather than extract from underserved communities.</i>	(2017) — <i>Liveright</i> — government-produced segregation - Sheller, Mimi, <i>Mobility Justice, Ch. 1</i> (2018) — Verso — <i>the right to move</i> - Klinenberg, Eric, <i>Palaces for the People, Introduction & Ch. 1</i> (2018) — Crown — <i>social infrastructure as equity</i>	(2017) - Sheller, Mimi, <i>Mobility Justice, Ch. 1–3</i> (2018) - Klinenberg, Eric, <i>Palaces for the People, Introduction & Ch. 1</i> (2018) - Pulido, Laura, <i>Rethinking Environmental Racism</i> (2000) — <i>Annals of the Association of American Geographers</i> - Grengs, Joe, <i>Community-Based Transportation Planning for Public Transit</i> (2002) — <i>Journal of Planning Education and Research</i> - Gebhardt, Larry et al., <i>Sharing Economy and the City: Airbnb and Equity</i> (2017) — <i>Journal of Urban Affairs</i> - Eubanks, Virginia, <i>Automating Inequality, Ch. 1</i> (2018) — St. Martin's — <i>automated systems and the poor</i> - Ramiller, Alex, <i>Displacing the Displaced</i> (2022) — <i>Urban Affairs Review</i> — <i>Uber and equity</i>
9	Regulation as Design <i>Working with (and within) permitting, procurement, and municipal partnerships. Regulation as a product design constraint.</i>	- Bertaud, Alain, <i>Order Without Design, Ch. 3</i> (2018) — <i>land use regulation and urban outcomes</i> - Green, Ben, <i>The Smart Enough City, Ch. 3</i> (2019) — MIT Press — <i>civic tech and accountability</i> - Lydon, Mike & Garcia, Anthony, <i>Tactical Urbanism, Introduction & Ch. 1</i> (2015) — Island Press — <i>working within regulatory gaps</i>	- Bertaud, Alain, <i>Order Without Design, Ch. 3–4</i> (2018) - Green, Ben, <i>The Smart Enough City, Ch. 2–3</i> (2019) - Lydon, Mike & Garcia, Anthony, <i>Tactical Urbanism, Introduction & Ch. 1–2</i> (2015) - Selznick, Philip, <i>The Moral Commonwealth, Introduction</i> (1992) — UC Press — <i>institutions, law, and community</i> - Frug, Gerald, <i>City Making, Introduction & Ch. 1</i> (1999) — Princeton UP — <i>city governance and law</i> - Dagan, Hanoch & Heller, Michael, <i>The Choice Theory of Contracts, Ch. 1</i> (2017) — Cambridge UP — <i>optional — procurement theory</i> - Townsend, Anthony, <i>Smart Cities, Ch. 9 (Buggy, Brittle and Bugged)</i> (2013) — <i>what happens when smart tech fails in the field</i> - O'Kane, Josh, <i>Sideways: The City Google Couldn't Buy, Introduction</i> (2022) — Random House Canada — <i>Sidewalk Toronto primary account</i>
10	Civic Tech <i>311 systems, open data portals, participatory budgeting tools — government as a product platform.</i>	- Green, Ben, <i>The Smart Enough City, Ch. 2 & 5</i> (2019) — MIT Press — <i>civic tech done right and wrong</i> - Townsend, Anthony, <i>Smart Cities, Ch. 6–7</i> (2013) — <i>civic hacking and open government</i> - Lerner, Josh, <i>Everyone Counts: Could 'Participatory Budgeting' Change Democracy?, Ch. 1</i> (2014) — Cornell UP — <i>participatory budgeting origins</i>	- Green, Ben, <i>The Smart Enough City, Ch. 2, 4–5</i> (2019) - Townsend, Anthony, <i>Smart Cities, Ch. 6–7</i> (2013) - Lerner, Josh, <i>Everyone Counts, Ch. 1–2</i> (2014) — Cornell UP - Schrock, Andrew, <i>Civic Tech, Introduction</i> (2022) — MIT Press — <i>history and critique of civic hacking</i> - Fung, Archon, <i>Empowered Participation, Introduction & Ch. 1</i> (2004) — Princeton UP — <i>democratic participation in practice</i>

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			• Morozov, Evgeny, To Save Everything, Click Here, Introduction & Ch. 1 (2013) — <i>solutionism and civic tech's limits</i> • Tufekci, Zeynep, Twitter and Tear Gas, Introduction & Ch. 1 (2017) — Yale UP — <i>digital tools and collective action</i> • Peck, Jamie, Constructions of Neoliberal Reason, Introduction (2010) — Oxford UP — <i>ideological framing of civic tech</i>
11	Climate & Resilience <i>Products that address urban heat, flooding, and energy. What does net-zero city infrastructure look like?</i>	• Klinenberg, Eric, Palaces for the People, Ch. 5 (2018) — <i>social infrastructure and climate resilience</i> • Chakrabarti, Vishaan, A Country of Cities, Introduction & Ch. 1 (2013) — Metropolis Books — <i>density as climate solution</i> • Mostafavi, Mohsen & Doherty, Gareth, Ecological Urbanism, Introduction (2010) — Harvard GSD — <i>environmental urbanism frameworks</i>	• Klinenberg, Eric, Palaces for the People, Ch. 4–5 (2018) • Chakrabarti, Vishaan, A Country of Cities, Introduction & Ch. 1–2 (2013) • Mostafavi, Mohsen & Doherty, Gareth, Ecological Urbanism, Introduction & selected essays (2010) — Harvard GSD • Bulkeley, Harriet et al., Urban Sustainability Transitions, Introduction (2016) — Routledge — <i>sustainability and urban governance</i> • Romero Lankao, Patricia, Urban Areas and Climate Change, Ch. 1 (2010) — ISSC/UNESCO — <i>cities as producers and victims</i> • Plyer, Allison, Facts for Features: Hurricane Katrina Impact (2016) — The Data Center — <i>empirical urban resilience</i> • Vale, Lawrence & Campanella, Thomas, The Resilient City, Introduction (2005) — Oxford UP — <i>disaster, recovery, and urban identity</i> • Beatley, Timothy, Biophilic Cities, Introduction & Ch. 1 (2011) — Island Press — <i>nature-integrated urban design</i>
12	Product Pitch Workshop <i>Teams present early urban product concepts. Structured critique from peers and invited urban practitioners.</i>	• Osterwalder, Alexander & Pigneur, Yves, Business Model Generation, Ch. 1 (2010) — Wiley — <i>the business model canvas for products</i> • Blank, Steve, The Four Steps to the Epiphany, Introduction (2005) — <i>customer discovery for urban context</i>	• Osterwalder, Alexander & Pigneur, Yves, Business Model Generation, Ch. 1–2 (2010) • Blank, Steve, The Four Steps to the Epiphany, Introduction & Ch. 1 (2005) • Mazzucato, Mariana, The Entrepreneurial State, Introduction & Ch. 1 (2013) — Anthem — <i>public investment and urban innovation</i> • Hoffman, Patrick T., @arcktip Practitioner Framework: Pricing as Product Design (2026) — @arcktip — <i>how pricing shapes urban access</i>
13	Go-to-Market in Cities <i>City-by-city expansion strategy, pilot design, stakeholder mapping, and</i>	• Hoffman, Patrick T., @arcktip Practitioner Framework: Launching in Cities (2026) — @arcktip — <i>practitioner go-to-market framework</i> • Rogers, Everett, Diffusion of Innovations, Introduction & Ch.	• Hoffman, Patrick T., @arcktip Practitioner Framework: Launching in Cities (2026) — @arcktip • Rogers, Everett, Diffusion of Innovations, Introduction & Ch. 1–2 (1962/2003)

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	<i>navigating municipal procurement.</i>	1 (1962/2003) — <i>Free Press</i> — how innovations spread through systems - Moore, Geoffrey, <i>Crossing the Chasm, Introduction & Ch. 1</i> (1991) — <i>HarperBusiness</i> — crossing from early adopters to mainstream	- Moore, Geoffrey, <i>Crossing the Chasm, Introduction & Ch. 1</i> (1991) - Mazzucato, Mariana, <i>The Entrepreneurial State, Ch. 1–2</i> (2013) — <i>public investment and risk in urban markets</i> - Ruef, Martin, <i>The Entrepreneurial Group, Introduction</i> (2010) — <i>Princeton UP</i> — social networks and startup success - Sassen, Saskia, <i>The Global City, Ch. 1 & 5</i> (1991) — <i>Princeton UP</i> — city as market context - Goldsmith, Stephen & Crawford, Susan, <i>The Responsive City, Introduction & Ch. 1</i> (2014) — <i>Jossey-Bass</i> — government as platform partner - Townsend, Anthony, <i>Smart Cities, Ch. 10–11</i> (2013) — <i>city-scale deployment lessons</i>
14	Failure Cases <i>What went wrong: cautionary tales from WeWork, Sidewalk Toronto, and Bird's rapid collapse.</i>	- O'Kane, Josh, <i>Sideways: The City Google Couldn't Buy, Introduction & Ch. 1</i> (2022) — <i>Random House Canada</i> — <i>Sidewalk Toronto</i> account - Green, Ben, <i>The Smart Enough City, Ch. 6–7</i> (2019) — <i>MIT Press</i> — <i>when civic tech fails</i> - Brown, Eliot, <i>The Cult of We (WeWork), Introduction</i> (2021) — <i>Crown</i> — <i>urban office product failure at scale</i>	- O'Kane, Josh, <i>Sideways: The City Google Couldn't Buy, Introduction & Ch. 1–3</i> (2022) - Green, Ben, <i>The Smart Enough City, Ch. 6–7</i> (2019) - Brown, Eliot, <i>The Cult of We, Introduction & Ch. 1</i> (2021) - Flyvbjerg, Bent et al., <i>Megaprojects and Risk, Introduction & Ch. 1</i> (2003) — <i>Cambridge UP</i> — <i>why large urban projects fail</i> - Scott, James C., <i>Seeing Like a State, Ch. 4 (The High-Modernist City)</i> (1998) — <i>Brasília and Chandigarh as cautionary tales</i> - Townsend, Anthony, <i>Smart Cities, Ch. 9 (Buggy, Brittle and Bugged)</i> (2013) — <i>systemic failure in smart city tech</i> - Spicer, André, <i>Business Bullshit, Introduction & Ch. 1</i> (2017) — <i>Routledge</i> — <i>language of urban tech hype</i> - Wachsmuth, David, <i>The Platform Urban Form</i> (2020) — <i>Built Environment</i> — <i>taxonomy of platform failures</i>
15	Final Pitch Day <i>Teams present fully developed urban product concepts to a panel of city planners, practitioners, and investors.</i>	No assigned reading — final pitch day. Preparation and rehearsal only.	- No assigned reading — final pitch day. Graduate teams respond to external panel critique. - Hoffman, Patrick T., <i>@arcktip Synthesis Essay: What Cities Actually Want From Products</i> (2026) — <i>@arcktip</i> — optional synthesis essay for post-course reflection

Reading Notes

Titles in italic are assigned texts. 'Ch.' indicates specific chapters only. All readings available via university library or open access unless noted.

@arcktip sources (Hoffman) are primary practitioner readings — first-hand accounts from building products at Social Bicycles / JUMP, Superpedestrian, Google, and Meta, and from independent research. These anchor the course in lived professional experience.

Graduate students read all listed texts. Undergraduate students read the 1–3 anchor texts and come prepared to discuss them in depth.